

Chapter 12, Problem 41P

Problem

A balanced delta-connected load draws 5 kW at a power factor of 0.8 lagging. If the three-phase system has an effective line voltage of 400 V, find the line current.

Step-by-step solution

Step 1 of 2

$$S = \frac{P}{pf} = \frac{5 \text{ KW}}{0.8} = 6.25 \text{ KVA}$$

Step 2 of 2

$$\text{But, } S = \sqrt{3} V_L I_L$$

$$I_L = \frac{S}{\sqrt{3} V_L}$$

$$= \frac{6.25 \times 10^3}{\sqrt{3} \times 400}$$

$$= 9.021 \text{ A}$$